

Ruaridh Murdoch

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I'm a Design Engineering student at Imperial College London, ranked first in my year, with a strong interest in mechatronics, robotics and human-centred design. My main strength is CAD, particularly designing parts and assemblies that can be integrated and used in the real world, drawing from experience that ranges from research-led, user-centred design to technical projects involving rapid prototyping, coding and electronics. I'm looking for a placement where I can contribute to real products from concept through to build.

Education

Imperial College London

London, UK

MEng Design Engineering, Dyson School of Design Engineering

Oct 2024 — Jun 2028

- **Head of School Prize (Year 2):** awarded for the highest average result across the Design Engineering cohort
- Year 2: 77.9% · Year 1: 72.7% — first class in both years
- Relevant coursework: Physical Computing, Electronics: Signals & Systems, Solid Mechanics, Finite Element Analysis, Thermofluids, Product Design Engineering, Human-Centred Design Engineering, Computing: Applications

Projects

SwivelCut: Motorised Cardboard Cutting Tool

2026

Group of 4

- Identified through cross-school user research (11 interviews, 92-person survey) that Design & Technology students were held back by manual cutting tools that couldn't accurately reproduce their own designs, leaving cardboard prototypes that consistently fell short of what they'd drawn.
- Built a 2-joint motorised arm with swappable cutting and tracing heads, sized to cut cleanly across a full A3 sheet, with a sprung blade mount that reduced peak cutting force from 28 N to 8 N.
- Delivered a £299 retail-costed, fully compliant product and validated it through user testing with 18 Design & Technology students across Years 7 - 12.

Teach-by-Touch Robotic Arm

2025

Group of 2

- Built a 4-joint robotic arm that learns motion by being physically moved: record a sequence by hand, insert a cartridge, press play to repeat it at original speed.
- Developed the joints from scratch under a low-cost constraint, using servo-equivalent assemblies built from DC motors, 3D-printed gear reductions and potentiometer feedback, with motion driven by a four-state finite state machine and sequences recalled via swappable colour-identified cartridges.
- Diagnosed and resolved failures under load, fixing shoulder stall by doubling the gear reduction and gripper locking by adding bearings.

Self-Balancing Robot

2026

Group of 2

- Programmed the balance control system for a provided two-wheeled, inverted-pendulum robot, implementing a PID feedback loop on pitch angle from onboard accelerometer and gyroscope data on a Pyboard (MicroPython).
- Built a real-time beat-detection routine so the robot's balancing motion synchronises with music, combining control and audio signal processing on the same embedded system.

Prosthetic Running Blade

2026

- Designed and analysed a CFRP sprinting blade against three criteria (mass, fatigue, frequency); the initial design failed all three, with stress at 2,924 MPa against a 600 MPa limit.
- Used beam theory to predict the required geometry changes before re-simulating, cutting stress to 419 MPa and achieving a ~2x frequency increase, predicted to within 1.5% of the FEA result.

Designing for Comfort in Long-Haul Aircraft

2025

- Ran a 122-person passenger survey, benchmarked seat manufacturers and tore down an aircraft seat to find where current designs fail.
- Identified the cushion-passenger interface as the point most worth redesigning, then proposed a modular seat with adjustable geometry and separable soft goods to fit more body types and reduce waste.

Battery Liquid Cold Plate: CFD

2026

Group of 4

- Designed a liquid cooling plate for a 740 W lithium-ion battery pack, developing a parallel-serpentine channel layout to balance thermal uniformity against pressure drop.
- Diagnosed flow maldistribution in the first layout by inspection before simulating it, removed the short central channel and equalised path lengths.
- Iterated across five variables to optimise the design, accepting a pressure drop increase from 80.8 Pa to 297 Pa in exchange for a cell temperature spread of 1.9°C across 18 modelled cells; well within the 5°C limit for avoiding uneven battery ageing.

Rhythm Rush: Social Play System for Event Queues

2025

Group of 5

- Researched event queues through street interviews, surveys and public experiments, finding that people avoid talking to strangers unless a shared activity, like a game, makes the first move feel lower-risk.
- Built and tested a real-time body-tracking prototype in Python using Google's MediaPipe, refining it with user feedback to handle multiple people and different lighting conditions.

Skills

- **Electronics & embedded:** Arduino (C++), MicroPython, ESP32, circuit prototyping, sensor integration, DC / stepper / servo motor control, PID and finite-state-machine control
- **Mechanical & CAD:** Fusion 360, tolerance & fit, gear & mechanism design, DFM / DFA, 3D printing, rapid prototyping, technical drawing
- **Simulation & analysis:** FEA and CFD in Ansys (Mechanical, Fluent), MATLAB, hand-calculation validation
- **Software:** Python, C#, C++, JavaScript, Git, Unity
- **Design & research:** user research & interviews, survey design, journey & stakeholder mapping, benchmarking, lifecycle & materials analysis, Figma, UI design

Experience

Undergraduate Teaching Assistant

Oct 2025 — Jul 2026

Imperial College London

London, UK

- Mentored over 90 first-year Design Engineering students weekly across maths, mechanics, materials and computing.

AI Model Trainer

Jul 2025 — Present

DataAnnotation

Remote

- Evaluate and annotate AI model outputs across reasoning, mathematics and programming, identifying failure modes and edge cases to improve model accuracy.

Young Enterprise Company Programme: Palletable Products

2023 — 2024

Co-Founder

Glasgow, UK

- Helped set up and run a student enterprise that redesigned disposed wooden pallets into bird boxes, using machine tools and timber treatment to give the reclaimed material a second life.
- Won the Best Interview, Teamwork and Best Product or Service awards at the Glasgow finals.

Volunteer Bicycle Mechanic

2023 — 2024

Recycle-a-Bike

Stirling, UK

- Stripped, repaired and rebuilt donated bikes for community redistribution: drivetrains, brakes and frames.

Coding Club Mentor

2022 — 2023

The Glasgow Academy

Glasgow, UK

- Led coding sessions for secondary students, breaking down programming concepts and guiding problem-solving.

Awards

- **Duke of Edinburgh:** Silver Award (2022), currently completing Gold
- **British Physics Olympiad:** Senior Physics Challenge, Silver Award (2023)
- **UK Bebras Challenge:** Gold Award in 2021 and 2022, including Best in School and a Certificate of Distinction

Interests

- **Game jams:** won Duck Sauce Game Jam two years running, shipping playable original games as 48-hour Unity and C# builds.
- **Climbing & outdoors:** NIBAS / NICAS-certified lead climber, former Glasgow Youth Squad competitor, completed the Lakeland Mountain Marathon.
- **Sailing:** RYA Day Skipper, sailed Scottish coastal waters independently.